



Telco Customer Churn Analysis – Detailed Report

Executive Summary

Our dataset contains **7,043 customers**, of which **26.5% (1,869 customers)** have churned. Analysis across service categories reveals strong correlations between churn and lack of support/protection services, internet type, and contract/payment methods.

Key Findings

1. Internet Service Type

- **Fiber optic users:** ~42% churn rate (highest among all groups).
- **DSL users:** ~19% churn rate.
- **No internet service:** ~8% churn rate.

👉 Fiber optic customers are nearly **2× more likely** to churn compared to DSL.

2. Support & Protection Services

- **Online Security:** Customers without it churn at **42%**, vs. **15%** with it.
- **Online Backup:** No backup → **39% churn**, with backup → **21% churn**.
- **Tech Support:** No support → **42% churn**, with support → **15% churn**.

👉 Lack of these services consistently increases churn risk by **20–25 percentage points**.

3. Streaming Services

- Streaming TV/Movies show **no strong impact** on churn (both groups ~35% churn).

👉 Entertainment services are not retention drivers.

4. Phone Service & Multiple Lines

- Phone service customers churn at **26%**, vs. **25%** without phone service.
- Multiple lines show negligible difference.

👉 These features do not significantly influence churn.

5. Contracts & Billing

- **Month-to-month contracts:** ~43% churn.
- **One-year contracts:** ~11% churn.
- **Two-year contracts:** ~3% churn.
- **Electronic check payments:** ~45% churn, vs. ~15% for credit card/bank transfer.

👉 Contract length and payment method are **critical churn predictors**.

Recommendations

1. Retention Campaigns for Fiber Optic Customers

- Investigate dissatisfaction drivers (pricing, service reliability).
- Offer loyalty discounts or service quality guarantees.
- Targeted retention could reduce churn by **10–15%** in this segment.

2. Bundle Support Services

- Promote packages including Online Security, Backup, and Tech Support.
- Customers with these services churn at **~15–21%**, vs. **~39–42%** without.
- Bundling could cut churn by **half** in vulnerable groups.

3. Contract Incentives

- Encourage one- and two-year contracts with discounts or rewards.
- Moving customers from month-to-month to annual contracts could reduce churn by **30 percentage points**.

4. Payment Method Optimization

- Incentivize auto-pay via credit card/bank transfer.
- Transitioning customers from electronic checks could reduce churn by **~30 percentage points**.

5. Customer Education

- Communicate benefits of bundled services and long-term contracts.
 - Use targeted messaging for high-risk groups (fiber optic, month-to-month, electronic check users).
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Business Impact Projection

- If **20% of fiber optic customers** adopt bundled support services and switch to annual contracts:
 - Potential churn reduction: **~400 customers retained annually**.
 - Estimated revenue retention: **\$400K–\$600K per year** (assuming \$1,000–\$1,500 average annual revenue per customer).